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Advertisement call of *Scinax camposseabrai* (Bokermann, 1968) (Anura: Hylidae), with comments on the call of three species of the *Scinax ruber* clade

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Scinax camposseabrai was allocated into the *Scinax ruber* clade by Caramaschi & Cardoso (2006) by overall similarities as snout not pointed, breeding in open areas, and an advertisement calls with multipulsed notes. This assumption about the call was based solely on an onomatopoeia provided by Bokermann (1968). Herein we provide a formal description of the advertisement call of *S. camposseabrai* and compare it with described calls of other *S. ruber* clade species. Additionally, we provide descriptions of the advertisement calls of three sympatric species of the *S. ruber* clade: *S. eurydice* (Bokermann), *S. pachycrus* (Miranda-Ribeiro) and *S.* cf. *x-signatus*.

Except for one individual of *Scinax eurydice* recorded from the municipality of Uruçuca (14°35′34″S, 39°17′2″W), state of Bahia, Brazil, all other individuals were recorded from Floresta Nacional Contendas do Sincorá (FNCS), municipality of Contendas do Sincorá (13°45'46"S, 41°02'27"W), state of Bahia, Brazil. The FNCS is a conservation unit in the Caatinga biome, located about 90 km southwest from municipality of Maracás, the type locality of *S. camposseabrai* and *S. eurydice*. Collected specimens are housed at Coleção Zoológica da Universidade Estadual do Sudoeste da Bahia (MHNJCH), State of Bahia, Brazil (permits # 35068). All recordings were made with Marantz PMD 660 recorder (sampling frequency 44.1 kHz and 16 bit) and Sennheiser ME 66 shotgun microphone. For the acoustic analysis we used Raven Pro 1.4 with the following settings: window type = Hann, window size = 1024 points, frequency grid spacing = 43.1 Hz and overlap = 50 %. Temporal parameters were measured from the waveform, whereas spectral parameters were measured from the spectrogram. We use the mechanistic definition of 'note' as a unit of sound produced by a single cycle of air through the larynx (McLister *et al.* 1995). We discriminated the notes by observing the contraction of trunk muscles and vocal sac expansion. We follow the definition of 'pulse' (emphasized energetic impulses of a 'note') according to Duellman & Trueb (1994), and the definition of 'pulse period' according to Littlejohn (2001).

Scinax camposseabrai

We recorded the advertisement calls of 13 males of *Scinax camposseabrai* on January 26th, May 30th and 31st, and December 9th 2013. One recorded male was collected (MHNJCH 568). Males of *S. camposseabrai* were found calling in open areas, floating or gently touching the bottom on shallow water (1–2 cm deep) at the margins of temporary ponds, except for one male that called from a leaf at 1.6 m above the ground (fig. 1A) and 4 m far from the margin. It is known that *S. camposseabrai* can inhabit epiphytic bromeliads, and males can be found calling floating in the middle of large rainwater pools or on low bushes and grasses at pond edges (Bokermann 1968; Caramaschi & Cardoso 2006; Cândido *et al.* 2012).

The advertisement call of *Scinax camposseabrai* is composed by a single type of note with internal amplitude modulation (multipulsed note) (fig. 1B–E). The first and last pulses have lower amplitude than the intermediate pulses (fig. 1C–D). Pulses are separated by 100% amplitude modulation (fig. 1E), and are sharp shaped, with rise time shorter than fall time. They also have frequency modulation (fig. 1E). The notes are emitted at high rates of 709 to 1150 notes per hour (mean = 998 ± 94, N = 63) in sequences of 6 to 156 notes (mean = 50.66 ± 34.86, N = 29 sequences) separated by very regular intervals (Table 1). These sequences lasted from 2.01 to 57.14 s (mean =

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19.33 ± 12.94, N = 29 sequences). The beginning note of note sequences has 2 to 9 pulses (mean = 5.93 ± 1.98, N = 27 sequences), the center note has 6 to 11 pulses (mean = 8.41 ± 1.55, N = 27 sequences), and the ending note has 4 to 11 pulses (mean = 7.65 ± 1.69, N = 23 sequences) (see Table 1 for overall mean and range of pulses per note). The mean number of pulses of initial notes was different from the number of pulses of center notes ($t = -4.09$, $p < 0.01$) and of ending notes ($t = -3.28$, $p < 0.01$). We found no difference between the number of pulses of the center notes and of the ending notes ($t = 1.64$, $p = 0.11$). Temperature was positively correlated with mean note emission rate ($r = 0.88$, $r^2 = 0.77$, $p < 0.01$, N = 13 males) and with mean pulse rate ($r = 0.88$, $r^2 = 0.77$, $p < 0.01$, N = 13 males).

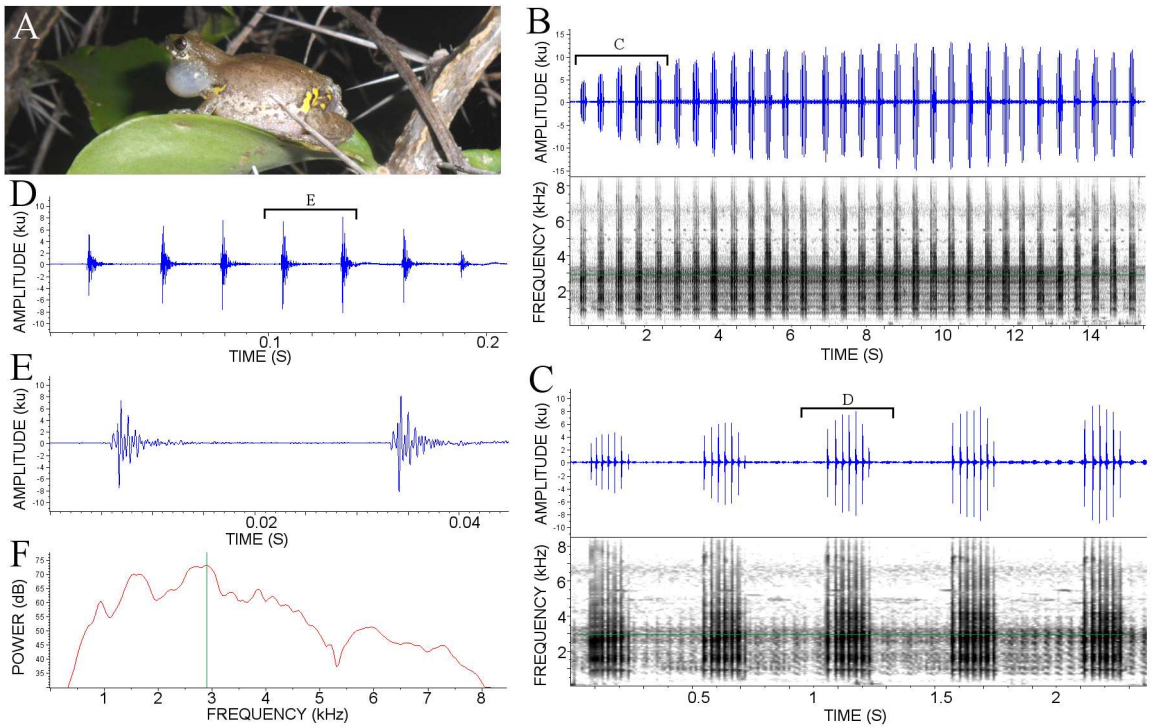


FIGURE 1. A recorded male of *Scinax camposseabrai* and the sonogram of its advertisement call, from Contendas do Sincorá, state of Bahia, Brazil. (A) Male of *S. camposseabrai* calling from a leaf of *Pereskia zehntneri* on May 30th 2013, air temperature 23 ºC, SVL 35.9 mm, unvouchered. (B) Waveform (above) and spectrogram (below) of a sequence of notes. (C) Waveform (above) and spectrogram (below) of the first five notes of the sequence. (D) Waveform of one note. (E) Waveform showing the shape of two pulses of the note. (F) Power spectrum of the call. Amplitude in waveforms is given in a relative unit. The green lines in the spectrograms and in the power spectra indicate the dominant frequency.

The frequency spectrum of *Scinax camposseabrai* advertisement call has three main peaks (bands) of energy between 0.7 and 3 kHz (fig. 1F). The first peak ranges from 732.1 to 1205.9 Hz (mean = 849.1 ± 122.6, N = 102 notes). The second peak ranges from 1076.7 to 1593.8 Hz (mean = 1189.1 ± 69.3, N = 102 notes). The third peak ranges from 2196.4 to 2953.1 Hz (mean = 2434.1 ± 187.3, N = 104 notes). The dominant frequency was always in the first peak in ten males; in the third peak in two males; and one male emitted some notes with dominant frequency in the first peak, and other notes with dominant frequency in the third peak. None of the frequency peaks were correlated with SVL (1st, 2nd, 3rd peak and dominant frequency, respectively: $r = -0.08$, $p = 0.78$; $r = -0.02$, $p = 0.94$; $r = 0.03$, $p = 0.92$; $r = 0.03$, $p = 0.93$).

Comparisons between the advertisement call of *Scinax camposseabrai* and other *S. ruber* clade advertisement calls

When the dominant frequency is in the first peak, we can distinguish the call of *S. camposseabrai* from those of *S. alter* (Lutz), *S. auratus* (Wied-Neuwied), *S. cabralensis* Drummond, Baêta & Pires, *S. caldarum* (Lutz), *S. cretatus* Nunes & Pombal, *S. crospedospilus* (Lutz), *S. curicica* Pugliese, Pombal & Sazima, *S. cuspidatus* (Lutz), *S. duartei*

(Lutz), *S. fuscomarginatus* (Lutz), *S. garbei* (Miranda-Ribeiro), *S. granulatus* (Peters), *S. hayii* (Barbour), *S. imbegue* Nunes, Kwet & Pombal, *S. juncae* Nunes & Pombal, *S. maracaya* (Cardoso & Sazima), *S. montivagus* Juncá, Napoli, Nunes, Mercês & Abreu, *S. nebulosus* (Spix), *S. pachycrus*, *S. perereca* Pombal, Haddad & Kasahara, *S. pinima* (Bokermann & Sazima), *S. rogerioi* Pugliese, Baêta & Pombal, *S. rupestris* Araujo-Vieira, Brandão & Faria, *S. similis* (Cochran), *S. squalirostris* (Lutz), *S. tigrinus* Nunes, Carvalho & Pereira, *S. tymbamirim* Nunes, Kwet & Pombal and *S. uruguayus* (Schmidt) (Table 1).

When the dominant frequency is in the third peak, we can distinguish it from the calls of *S. acuminatus* (Cope), *S. auratus* (call described by Nunes *et al.* 2007), *S. cabralensis*, *S. cretatus*, *S. crospedospilus* (call described by Bevier *et al.* 2008), *S. fuscomarginatus*, *S. fuscovarius* (Lutz), *S. garbei*, *S. granulatus*, *S. hayii*, *S. juncae*, *S. montivagus*, *S. nasicus* (Cope), *S. pachycrus*, *S. perereca* (call described by Pombal *et al.* 1995a), *S. pinima*, *S. rupestris*, *S. similis*, *S. squalirostris*, *S. tymbamirim*, *S. uruguayus*, *S. x-signatus* (Spix) and *S.* cf. *x-signatus* (Table 1).

We can distinguish the advertisement call of *S. camposseabrai* by the note durations, which are shorter than those of *S. acuminatus*, *S. alter*, *S. cabralensis*, *S. caldarum*, *S. crospedospilus*, *S. curicica*, *S. duartei*, *S. fuscomarginatus*, *S. garbei*, *S. granulatus*, *S. imbegue*, *S. maracaya*, *S. pachycrus*, *S. perereca*, *S. rogerioi*, *S. rupestris*, *S. squalirostris*, and *S. tymbamirim* (Table 1).

The interval between notes of the advertisement call of *S. camposseabrai* is shorter and less variable than those of *S. alter*, *S. auratus*, *S. cabralensis*, *S. crospedospilus*, *S. eurydice*, *S. fuscomarginatus*, *S. fuscovarius*, *S. garbei*, *S. granulatus*, *S. juncae*, *S. montivagus*, *S. nebulosus*, *S. pachycrus*, *S. perereca*, *S. rogerioi*, *S. similis*, *S. tigrinus*, and *S. x-signatus* (Table 1).

The advertisement call of *S. camposseabrai* is distinguished by the number of pulses per note from those of *S. acuminatus*, *S. alter*, *S. curicica*, *S. fuscomarginatus*, *S. garbei*, *S. hayii*, *S. pachycrus*, *S. perereca*, and *S. tymbamirim* (advertisement call with more pulses per note in these species), and from the call of *S. eurydice*, which notes never have more than 2 or 3 pulses, while in the call of *S. camposseabrai* only the initial note of a sequence may present 2 or 3 pulses and it occurs very infrequently (n = 1 and n = 2 respectively) (Table 1).

The pulse period of the advertisement call of *Scinax camposseabrai* is longer than those of *S. acuminatus*, *S. alter*, *S. cabralensis*, *S. caldarum*, *S. fuscomarginatus*, *S. hayii*, *S. nasicus*, *S. pachycrus*, and *S. perereca*; and it is shorter than those of *S. crospedospilus*, *S. eurydice* (present study), *S. juncae*, and *S. rogerioi* (Table 1).

Scinax eurydice

We recorded one male of *S. eurydice* on May 31st 2013 (unvouchered) and four males on January 18th and 19th 2015 (MHNJCH 1001–1002, two unvouchered) at the FNCS, state of Bahia, Brazil. Also, we recorded and collected one male of *S. eurydice* (MHNJCH 870), on February 28th 2014 at the municipality of Uruçuca, state of Bahia, Brazil. The males from FNCS called from the ground, at the margins of a temporary pond or perched up to 1.2 m high on the marginal or emergent vegetation. The male from Uruçuca called at a flooded area, from inside a terrestrial bromeliad, where one calling male of *S. pachycrus* was observed.

The advertisement call of *Scinax eurydice* is composed by a single type of note, with two pulses (fig. 2A, Table 1). It was emitted at a rate of 157 to 957 notes per hour (mean = 541 ± 159, N = 54). The two pulses of the note are separated by an amplitude modulation of about 99% (fig. 2A). Both pulses present frequency and amplitude modulation. Each pulse has seven or eight peaks of amplitude (fig. 2A). Pulse fall time is shorter than rise time, and the second pulse has more energy than the first one (fig. 2A). Most of the energy of the note is comprised between 0.8 kHz and 3 kHz. The frequency spectrum has two emphasized bands with similar amounts of energy, the lower band is around 0.85 kHz and the higher one is around 2.5 kHz (fig. 2A). The dominant frequency may be either one but it is more frequently the lower band (Table 1). SVL was inversely correlated with the lower frequency band ($r = -0.92$, $p < 0.05$). Air temperature was correlated with emission rate of pulses ($r = 0.97$, $p < 0.05$).

The advertisement call of *S. eurydice* was previously described by Pombal *et al.* (1995b) from Conceição da Barra, state of Espírito Santo and by Magrini *et al.* (2011) from Atibaia, state of São Paulo, both states in Brazil. Although there are minor differences between the dominant frequency, note duration and pulse period, the main difference between the calls described here and the previous descriptions is concerned to the number of pulses (which Pombal *et al.* 1995b called notes) per note: three in the calls from Espírito Santo and São Paulo, and only two in the calls that we analyzed from Bahia (fig. 2A, Table 1). Although this suggests acoustic distinction between the northeastern and southeastern populations, acoustic data of *S. eurydice* from other states of northeastern region are still lacking.

TABLE 1. Characteristics of advertisement call of some Brazilian *Scinax ruber* clade species. Values are presented as mean ± standard deviation (min–max) and number of notes analyzed. For dominant frequency, instead of discrete values, some authors provide ranges, which are showed here as numbers separated by a dash and without parenthesis. We used the pulse duration or pulse interval for comparisons, when the references did not provide pulse period, and we could not assess it from the sonograms.

Species	Dominant Frequency (kHz)	Note duration (s)	Interval between notes (s)	Number of pulses per note (s)	Pulse period (s)	References
Scinax acuminatus	0.83–1.56	0.67 (0.37–1.42)	-	32.6 (20–69)	0.01[a]	Magrini *et al.* 2011
S. alter	1.3–4.6	0.86 (0.38–2.07)	0.38–4.69	64.96 (29–152)	0.013[a]	Pombal *et al.* 1995b, Nunes *et al.* 2012
S. auratus	4.19 (4.00–4.35)	0.19 (0.15–0.22)	0.91–3.98	9*	0.009–0.026[1]	Nunes *et al.* 2007
	2.33 ± 0.20	0.14 ± 0.004	2.87[a]	5*	0.028[a]	Bevier *et al.* 2008
S. cabralensis	3.99 (3.70–4.22)	0.64 (0.31–1.01)	1.99 (0.73–5.73)	6–19	0.006*	Drummond *et al.* 2007
S. caldarum	2.41–2.61	0.26	-	7–13	0.01[2]	Pugliese *et al.* 2004
S. camposseabrai	0.81 ± 0.05 (0.73–0.90) N = 89 or 2.78 ± 0.1 (2.50–2.95) N = 15	0.15 ± 0.03 (0.02–0.20) N = 157	0.20 ± 0.05 (0.15–0.36) N = 63	8.47 ± 1.9 (2–11) N = 157	0.021 ± 0.004 (0.016–0.035) N = 157	Present study
S. cretatus	1.38	0.20 (0.19–0.23)	0.25–7.46	7–8	0.03 (0.02–0.05)[1]	Nunes & Pombal 2011
S. crospedospilus	1.41 ± 0.01	0.28 ± 0.01	0.69[a]	7*	0.04[a]	Bevier *et al.* 2008
	1.07–1.66 or 2.59–4.01	0.36 (0.28–0.44)	-	5–7	0.04[a]	Magrini *et al.* 2011
S. curicica	2.58–3.63	1.72 (0.76–4.5)	-	29–43	0.02	Pugliese *et al.* 2004
S. cuspidatus	2.3–4.0	0.14 (0.12–0.15)	-	5–6	0.023*	Pombal *et al.* 1995b
S. duartei	2.15–2.85	0.33	-	4–8	0.03[2]	Pugliese *et al.* 2004
S. eurydice	1–3.8	0.10 (0.09–0.11)	-	3	0.02–0.04[b]	Pombal *et al.* 1995b
	1.18–2.62	0.11	0.65[a]	3	0.02[a]	Magrini *et al.* 2011
	0.85 ± 0.03 (0.82–0.90) N = 54 or 2.65 ± 0.02 (2.63–2.67) N = 7	0.07 ± 0.00 (0.06–0.08) N = 61	0.68 ± 0.34 (0.31–2.21) N = 54	2 N = 61	0.045 ± 0.003 (0.040–0.051) N = 61	Present study
S. fuscomarginatus	3.82 (3.07–4.2)	0.54 (0.33–0.62)	3.11 (0.58–19.59)	95.2 (64–120)	0.006[b]	Toledo & Haddad 2005
S. fuscovarius	0.8–3.6	0.17–0.22	-	8–10	0.02[1]	Pombal *et al.* 1995b
	0.97 ± 0.07	0.21 ± 0.01	2.83[a]	9*	0.02[a]	Bevier *et al.* 2008
S. garbei	3.7 (3.4–3.9)	0.95 (0.3–1.5)	1.3	53 (16–83)	0.02[a]	De La Riva *et al.*1994

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TABLE 1. (Continued)

Species	Dominant Frequency (kHz)	Note duration (s)	Interval between notes (s)	Number of pulses per note (s)	Pulse period (s)	References
S. granulatus	1.28 (1.12–1.57)	0.24 (0.21–0.29)	1.26 (0.8–2.0)	9.2 (7–11)	0.02[a]	Conte *et al.* 2010
S. hayii	1.08–2.06	0.23 (0.19–0.27)	-	14.2 (13–17)	0.01[a]	Magrini *et al.* 2011
S. imbegue	1.2–5.0	0.23–0.38	-	9–14	0.029[a]	Pombal *et al.* 1995b, Nunes *et al.* 2012
S. juncae	3.66 (3.57–3.79)	0.22 (0.17–0.24)	1.39 (1.04–2.34)	4–5	0.05[b]	Nunes & Pombal 2010
S. maracaya	1–3.5	0.28	-	11–12	-	Cardoso & Sazima 1980
S. montivagus	1.70 (1.54–1.84)	0.20 (0.14–0.24)	2.22 (0.70–6.86)	12.65 (9–15)	0.016–0.019[a]	Juncá *et al.* 2015
S. nasicus	0.97 (0.85–1.11)	0.06 (0.04–0.1)	0.1*	5 (4–6)	0.01[a]	De La Riva *et al.*1994
S. nebulosus	2.37 (2.22–2.56)	0.24 (0.2–0.3)	2.4 (1–25)[a]	13 (11–17)	0.02[a]	De La Riva *et al.*1994
S. pachycrus	1.83 or 4.04	0.30 ± 0.03	1.36 ± 0.25	23.75 (17–32)	0.011*	Carneiro *et al.* 2004
	4.61 ± 0.18 (4.13–4.78) N = 16 or 2.10 ± 0.02 (2.07–2.11) N = 3	0.33 ± 0.11 (0.14–0.53) N = 19	0.89 ± 0.31 (0.53–1.81) N = 17	30.11 ± 9.54 (13–48) N = 19	0.011 ± 0.000 (0.011–0.012) N = 19	Present study
S. perereca	1.3–2.0	0.28–0.35	-	21*	0.01*	Pombal *et al.* 1995a
	2.50 ± 0.27	0.25 ± 0.01	4.56[a]	17*	0.01[a]	Bevier *et al.* 2008
S. pinima	3.3	0.1	0.2	-	-	Bokermann & Sazima, 1973
S. rogerioi	1.38–3.19	0.46 (0.27–0.70)	1.19	6–12	0.05[b]	Pugliese *et al.* 2009
S. rupestris	2.08 (2.07–2.22)	0.35 ± 0.03 (0.29–0.42)	-	11.06 ± 1.57 (8–14)	0.032[b]	Araujo-Vieira *et al.* 2015 [3]
S. similis	3.68 (3.19–4.13)	0.18 (0.13–0.22)	0.4–2.4	7 (4–10)	0.029 (0.022–0.045)	Bilate & Lack 2011
S. squalirostris	3.9–4.6	0.6 (0.54–0.68)	-	11–13	-	Pombal *et al.* 1995b
	4.5 ± 0.4	0.46 ± 0.15	-	11 ± 9.37	-	Faria *et al.* 2013
S. tigrinus	2.93–3.27	0.08–0.34	0.6–6.53	4–15	0.023*	Nunes *et al.* 2010
S. tymbamirim	4.5	0.5*	-	19*	0.025[a]	Nunes *et al.* 2012
S. uruguayus	4.15 (3.8–4.3)	0.02	0.24	7–10	-	Kolenc *et al.* 2003
S. x-signatus	1.25	0.2	0.9	8*	0.025[a]	Rivero 1969
S. cf. *x-signatus*	1.01 ± 0.02 (0.98–1.03) N = 17	0.14 ± 0.02 (0.11–0.17) N = 17	0.63 ± 0.25 (0.34–1.15) N = 15	8.82 ± 1.01 (7–10) N = 17	0.018 ± 0.001 (0.016–0.019) N = 17	Present study

* Values measured from sonograms provided in the references; [a] Calculated from the emission rate provided ; [b] The sum of the mean of pulse duration and mean of pulse interval; [1] Only pulse duration; [2] Only pulse interval; [3] We believe that what Araujo-Vieira *et al.* 2015 considered as 'call' is in fact one 'note' *sensu* McLister *et al.* (1995), therefore, what they named 'call' is herein treated as 'note' and what they named 'notes', are herein treated as 'pulses'.

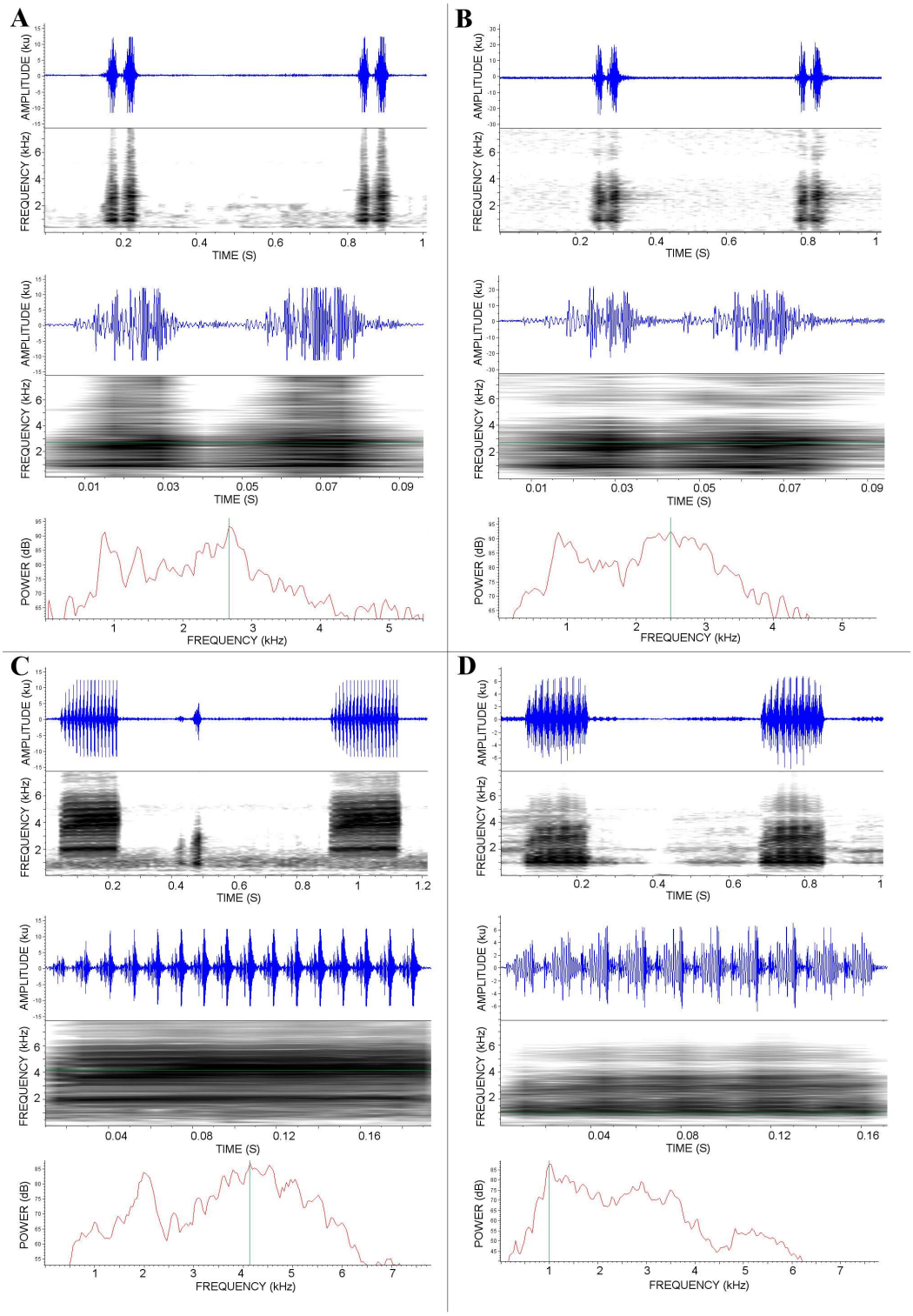


FIGURE 2. Waveform and spectrogram of two consecutive notes (above), waveform, spectrogram and power spectrum of one note (below) of the advertisement call of (A) *Scinax eurydice*, Contendas do Sincorá, state of Bahia, Brazil, recorded on January 18th 2015, air temperature 26 °C, SVL 44.2 mm, MHNJCH 1001; (B) *Scinax castroviejoi*, Santa Cruz, Bolivia, recorded by Ignacio De la Riva and deposited at www.fonozoo.com (Museo Nacional de Ciencias Naturales, Madrid); (C) *Scinax pachycrus*, Contendas do Sincorá, state of Bahia, Brazil, recorded on January 18th 2015, air temperature 25.8ºC, SVL 32.1 mm, unvouchered; (D) *Scinax* cf. *x-signatus*, Contendas do Sincorá, state of Bahia, Brazil, recorded on February 24th 2015, air temperature 22ºC, SVL 38.4 mm, MHNJCH 1014. Amplitude in waveforms is given in a relative unit. The green lines in the spectrograms and in the power spectra indicate the dominant frequency.

Scinax eurydice advertisement call is similar to the call of *S. castroviejoi* De la Riva, according to described data and spectrograms and waveforms; see De la Riva (1993) and De la Riva *et al.* (1994). *Scinax castroviejoi* is a species of the *S. ruber* clade from the valleys of the Bolivian Andes. Under the permission of Dr. Ignacio De la Riva, we analyzed one of his recording from Laguna Volcan, Santa Cruz, Bolivia, deposited in the sound library of Museo Nacional de Ciencias Naturales, Madrid (www.fonozoo.com). *Scinax castroviejoi* also emits a single type of note, with two pulses (which De la Riva 1993 called phases and De la Riva *et al.* 1994 called notes) of similar shape to that of *S. eurydice* (fig. 2B).

Means of note duration (0.07 s, N = 10), interval between notes (0.64 s, N = 9), and pulse period (0.04 s, N = 10) of *Scinax castroviejoi* advertisement call are very similar to the corresponding parameters of *S. eurydice* call. The range of frequencies is also very similar to that of *S. eurydice*, with two emphasized bands: one around 0.85 kHz (which De la Riva 1993 called fundamental frequency) and another around 2.5 kHz (fig. 2B). The only difference between the calls of these species is that the dominant frequency of *S. eurydice* is more common in the first band (0.85 kHz) whereas that of *S. castroviejoi* is more common in the second band (2.5 kHz). As *S. eurydice*, males of *S. castroviejoi* also call from the ground or from vegetation at the edge of ponds (De la Riva, 1993).

Scinax pachycrus

We recorded two males of *Scinax pachycrus*, one on April 1st 2013, and the other one on January 18th 2015. Males called from the ground, near the edge of the pond, or perched on emergent vegetation. The advertisement call is composed by a high-pitched multipulsed note, with 13 to 48 pulses (fig. 2C, Table 1). It was emitted at a rate of 170 to 681 notes per hour (mean = 337 ± 116, N = 18). The pulses have fall time shorter than rise time, and they are very close to each other, separated by an amplitude modulation of about 95% (fig. 2C). Each pulse presents frequency and amplitude modulation, with four crescent amplitude peaks; the last peak has 5 to 10 times the amplitude of the first one (fig. 2C). The most part of the energy in the spectrogram is found between 1.5 kHz and 6.5 kHz (fig. 2C). There are two more emphasized bands: one around 2.0 kHz and another around 4.6 kHz, and the dominant frequency can be either one (fig. 2C, Table 1).

Carneiro *et al.* (2004) recorded and analyzed the calls of six males from the municipality of Santa Teresinha, state of Bahia, Brazil about 206 km northeast from Contendas do Sincorá, state of Bahia, Brazil. Although the call described by Carneiro *et al.* (2004) presented lower frequencies our results are in general similar to their results (Table 1).

Scinax cf. *x-signatus*

Scinax x-signatus is a species of very confusing taxonomy, and it is likely a species complex (Rodrigues *et al.* 2010), with a broad geographic distribution. The description of advertisement calls from different populations under the name *S. x-signatus* may help to identify the existence of cryptic taxonomic entities.

We recorded two males of *Scinax* cf. *x*-signatus, one on May 31st 2013 (unvochered specimen), and another on February 24th 2015 (MHNJCH 1014). They called from the ground, near the edge of temporary ponds. The advertisement call of *S.* cf. *x-signatus* is a low-pitched multipulsed note, with 7 to 10 amplitude modulated pulses (fig. 2D, Table 1), emitted at a rate of 272 to 809 notes per hour (mean = 517 ± 170, N = 15). The pulses have four crescent peaks of amplitude, and the fall time shorter than the rise time. The frequency spectrum ranges from about 0.7 to 4.5 kHz, and the dominant frequency is around 1.0 kHz (fig. 2D, Table 1). *Scinax* cf. *x-signatus* was the only species that did not switch the dominant frequency between different bands of the spectrum.

Rivero (1969) provided a brief description and spectrograms of the advertisement call of *Scinax ruber* and *S. x-signatus* from Venezuela. He attributed two different calls to *S. x-signatus*. The first, with pulse rate of 40 pulses per second, and dominant frequency of 1.25 kHz (see fig. 2 in Rivero 1969), resembles the call of *S.* cf. *x-signatus* herein described (fig. 2D, Table 1). However, the second call, with pulse rate of 80 pulses per second, and two additional short notes (see fig. 1 in Rivero 1969), is more similar to the call of *S. ruber* therein described (see fig. 3 in Rivero 1969), and to the call of *S. ruber* described by De la Riva *et al.* (1994). It is likely that Rivero (1969) has considered this call as another type of call of *S. x-signatus*, but it actually belongs to a *S. ruber*, as already noticed by De la Riva *et al.* (1994).

Final remarks

The calls herein described present many sidebands along a broad range of frequencies due to the amplitude and frequency modulations and periodicity of pulses. In some cases, two or more sidebands can have comparable quantities of energy and can alternate as the dominant frequency of the call. Magrini *et al.* (2011) also observed multiple emphasized frequencies, as well as intra- and inter-individual variation of dominant frequency in other *Scinax ruber* clade species. This characteristic seems to be very common in the species of this clade and was also observed in other species of the Dendropsophini Tribe (*sensu* Faivovich *et al.* 2005; Conte *et al.* 2010; Hepp *et al.* 2012).

Changes in dominant frequency are often related to social context (*e.g.* Zina & Haddad 2006; see also Wells 2007 for review). However, it may not always have an adaptative explanation, as it could be related to changes in other call properties as duration of notes and call intensity (Wells 2007). These changes also may be a consequence of other factors, for example: fluctuations of air temperature and humidity; dynamics of the physiology of the call production; sensitivity or condition of the recording equipment and analysis software; researcher bias, especially when two or more frequency bands of the species call have similar amounts of energy (*e.g.* Lingnau *et al.* 2008). Furthermore, Magrini *et al.* (2011) expected a taxonomic relevance of multiple bands in *Scinax* species and, therefore, recommended special attention in the description of spectral components. Indeed, we found that the only difference between the calls of *S. eurydice* and *S. castroviejoi* is the prevalence of the lower frequency band as dominant frequency in *S. eurydice*, and the prevalence of the higher frequency band as dominant frequency in *S. castroviejoi*. Taxonomic and phylogenetic studies may resolve if the similarities between their calls reflect homology or result from convergence.

Some characteristics of the advertisement call of *Scinax camposseabrai* (call composed by a single type of multipulsed note, emitted at high rates, with a broad spectrum of frequencies and multiple interchangeable emphasized bands) are presented by other species of the *S. ruber* clade (Pombal *et al.* 1995b; Bevier *et al*. 2008; Magrini *et al.* 2011; see Table 1 for other references) demonstrating the importance of the description of advertisement calls as an relevant tool in the study of the relationships between the species.

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